Greenbrier Invitational - Repaired practice edition - Answer key

Division C | Circuit Lab | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

1. (2 points) Authored replacement for source item Question 1: A negative charge's force is…

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

A. Opposite the electric field

B. Always with the field

C. Always zero

D. Independent of its charge

Correct: A

6. (2 points) Authored replacement for source item Question 6: Define "Potential" in the context of Circuit Lab.

answer: Electric potential energy per charge. (2 points).

repair-case. (6 points) Authored written substitute for unavailable source material: Solve this case and explain your reasoning: Two point charges keep their values while separation triples. Predict force.

reasoning: Use F proportional to 1/r^2.
Replacing r with 3r multiplies the denominator by nine.
The new magnitude is F/9; attraction or repulsion is unchanged. (3 points).

conclusion: Direction depends on signs, not distance alone. (3 points).